

DANIEL D. JOHN

Controls & Simulation Engineer | Engineering Physics

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SUMMARY

Engineering physics student and simulation developer focused on control systems, vehicle/spacecraft dynamics, guidance-oriented algorithms, and real-time computational simulation. Experience building physics-based models from first principles, developing actuator-allocation and trajectory solvers, and validating behavior through numerical analysis and scenario testing.

TECHNICAL SKILLS

Controls & Simulation: control systems, spacecraft/vehicle dynamics, trajectory mechanics, actuator/thruster allocation, N-body & Keplerian orbital modeling, real-time simulation, numerical modeling, model validation

Programming & Tools: C# (10 years), MATLAB, Mathematica, JMP, Unity (9 years), CAD/CAM

Relevant Academic Focus: dynamics, controls, numerical methods, embedded computation, systems engineering

ENGINEERING EXPERIENCE

Independent Simulation Developer

2016–2026

- Developed real-time, physics-driven simulation systems for aerospace and vehicle dynamics applications.
- Designed rocket thruster-allocation algorithms for distributing control effort across actuator configurations.
- Implemented flight-dynamics and trajectory solvers using blended N-body and Keplerian orbital models to capture perturbations and long-arc motion.
- Built guidance-oriented target-selection scoring logic and numerical scenario tests for autonomous simulation behavior.
- Applied computational physics and model-validation methods to test system behavior, edge cases, and simulation fidelity.

Minnesota Army National Guard — Artillery Specialist

Jul 2023–Present

- Perform precision-oriented tasks requiring disciplined procedures, systems awareness, teamwork, and reliable execution in high-pressure environments.

SELECTED PROJECTS

Decel — Solar System & Spacecraft Dynamics Simulator

- Built a spacecraft simulation platform centered on N-body orbital mechanics, spacecraft control, and trajectory planning.
- Developed physics-based computational methods for long-duration orbital motion and control-oriented simulation.

GNC Trajectory Visualization

- Developed tools for visualizing and comparing guidance, navigation, and control trajectories to support trajectory analysis.

ARA Rally Car — Driver & Mechanic

- Build and maintain rally vehicle systems; troubleshoot mechanical issues and perform field repairs under competitive conditions.

EDUCATION

University of Minnesota — Twin Cities & Duluth

Jun 2024–Present

Bachelor of Science, Engineering Physics Systems — EAC-ABET accredited program

Focus: dynamics, controls, computational methods, systems engineering, and applied physics.